

Question number	Answer	Additional guidance	Mark
7(a)	An explanation that makes reference to the following points: - because (both) parents could be {carriers of / heterozygous for} {the disorder / phenylketonuria} (1) - all possible genotypes (of offspring) {shown in diagram / stated} (1) - therefore there is the possibility that the unborn child could be {homozygous recessive / have both recessive alleles} (1) - credit a reason for parents wanting to have the test (1)	NB must be an attempt at genetic diagram for full marks to be awarded ACCEPT from a genetic diagram DO NOT ACCEPT different letters used for the different alleles unless stated sex linkage CE from mp 1 for mps 2, 3 and 4 ACCEPT if indicated on the diagram ACCEPT if indicated on the diagram e.g. do not want to risk having a child with PKU want time to prepare for a child with PKU because there is a history of PKU in the family so there is risk of the child having PKU they know that {one / both} of them are carriers so there is a risk of the child having PKU so that an unaffected embryo can be selected DO NOT ACCEPT abortion / termination / other post-natal decisions / choosing to have a child or not IGNORE making an informed decision	(4)

Question number	Answer	Mark
*7(b)	Blood tests • most women would have a blood test • similar percentage for all four disorders • probably because it would not harm the unborn child • but some women may not want to find out if they have a genetic disorder • it would not take into account the genetics of the father Amniocentesis • about two thirds of the women said that they would have amniocentesis • similar percentage for all four disorders • fewer women said yes to this question than agreed to have a blood test • probably because they wanted to find out if their unborn child was affected • so they could make a decision on whether to keep the unborn child or not • the more debilitating the disease the more women prepared to take the risk of amniocentesis causing an abortion • amniocentesis can cause abortion of the unborn child • and some women must have felt that this was {too great a risk / unethical} • amniocentesis can give false negative and false positive results Termination • fewer women said that they would have a termination than were prepared to have the other procedures • fewer said yes to disorder B and C than A and D because B and C are treatable • more women said that they would have a termination if disorder A was diagnosed as this can be lethal • the more debilitating the disease the more likely the women were to have a termination because {do not want their child to suffer / cannot afford the healthcare} • some women may say no to a termination for religious or ethical reasons	(6)

Level 1 : description of the data with an attempt to discuss why responses are different

1 mark = some of the data described

2 marks = some of the data described with one discussion point made

Level 2 : discussion of some of the data

3 marks = two discussion points made that may be limited to one question

4 marks = two discussion points made about two of the questions

Level 3 : discussion of data

5 marks = responses to the three questions for all the procedures discussed

6 marks = plus at least one reason given for the different responses to a question depending on the disorder